

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 08

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>2 marks for each point (partial credit for a partly correct point).</i> • Normal diagonal mesh closes as the net fills / stretches so small fish cannot escape; square-mesh panels do not close so juveniles escape to grow and breed; TEDs let turtles out; larger mesh sizes generally (2) • Minimum size lets fish reach breeding age before capture (eg UK lobster minimum carapace above breeding size, cod 35 cm); maximum size / V-notching egg-carrying females protects large, highly fecund breeding adults (Hebridean lobsters over 145 mm, Nassau grouper) (2) • NTZs (Lundy, Lamlash Bay, Chagos, Great Barrier Reef) protect breeding populations and habitats; surplus young spill over and increase surrounding catches; damaged communities recover (2) • Closed seasons, ideally in the breeding season, let fish grow and breed; fishermen may oppose because shoaling breeders are easy to catch (2)	8
01.2	<i>1 mark for each point.</i> • Pelagic species form single-species shoals so fishing stops when the quota is reached (1) • In mixed fisheries many species are caught together; once one species' quota is reached it must be discarded (dead) while fishing for others continues (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 344-348; AQA spec 3.5.2.2.	
02.1	<i>1 mark for each point.</i> • Coordinating global data on biodiversity conservation (Red List) (1) • Increasing understanding of the importance of biodiversity (1) • Deploying nature-based solutions to global challenges in climate, food and sustainable development (1)	3

Question	Marking guidance	Marks
02.2	<i>1 mark for each point.</i> - Decline = $(12\,000 - 2100) / 12\,000 \times 100 = 82.5\%$ (1) - Decline over 80% meets the Critically Endangered criterion even though habitat area (400 km^2) and adults (2100) only meet Endangered (1) - A species is placed in the highest category for which it meets any one criterion, so Critically Endangered (1)	3
02.3	<i>1 mark for each point.</i> - Categorising a species requires a great deal of research (population, range, trends) and resources; about 85 000 of millions of species assessed (1) - Key species in habitats under particular threat (eg lemurs indicating the state of Madagascar's forests) (1) - Evolutionarily unique species / EDGE species; endemic species; keystone species; flagship species; species with fragmented populations (1)	3
02.4	<i>1 mark.</i> Iberian lynx: Critically Endangered to Endangered after captive breeding and protection of two habitat areas (population trebled in 15 years); Fregate Island beetle after rat eradication (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 26-27; AQA spec 3.1.2.3.1.	
03.1	<i>1 mark for the correct answer.</i> $2.7 - 2.45 = 0.25$ billion = 250 million years (1)	1
03.2	<i>1 mark for each point.</i> - For millions of years all the oxygen produced reacted with (dissolved) iron in the oceans (1) - Only when all the iron had reacted did surplus dissolved oxygen build up in the oceans and then escape into the atmosphere (1)	2
03.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> - Oxygen in the atmosphere absorbed ultraviolet light (1) - A dynamic equilibrium of reactions between O, O₂ and O₃ (the ozone layer) developed (1) - Less UV reached the surface so organisms were no longer restricted to water (which had absorbed the UV) (1) - Aerobic respiration became possible so more complex / larger organisms (plants and animals) evolved; many anaerobic Archaea and bacteria died out (1)	3
03.4	<i>1 mark for each point.</i> - Photoautotrophs absorbed carbon dioxide, a greenhouse gas, and some of the carbon was stored in geological sediments such as carbonate rocks and fossil fuels (carbon sequestration) (1) - The lower CO₂ concentration reduced the retention of heat, offsetting the increase in the brightness of the Sun (about 10% per billion years) (1)	2
03.5	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> - Biogeochemical cycles developed so small amounts of nutrient elements could support life for long periods without depletion (1) - Transpiration by land plants returned water vapour to the atmosphere and increased rainfall in other areas (1) - Regulation of atmospheric composition (eg oxygen / CO₂ by photosynthesis and respiration) (1)	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 6-8; AQA spec 3.1.1.2.	
04.1	<i>1 mark for each point.</i> • No major temperature extremes; regular water supply with no pronounced dry season; deep fertile soils built by leaf litter and root binding; strong seasonality (1) • Deciduous canopy present spring to autumn shades the floor during the best growing temperatures (1) • Bluebells, snowdrops, lesser celandine grow and flower early in spring before the trees leaf (1) • Winter food shortage causes animals to hibernate (dormice), store food or migrate (1)	4
04.2	<i>1 mark for each point.</i> • Long time for species to colonise; some woodland plants colonise very slowly so their presence (ancient woodland indicators) shows the wood is old (1) • Natural woodland has trees of all ages with clearings where old trees died, giving varied light and dead wood habitats (1) • A new plantation has a single-age dense canopy with low light and few niches until trees start to die (over 200 years) (1)	3
04.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Clearance for farmland, plantations, urban development, roads, mineral extraction (1) • Fragmentation isolating remaining woods so populations cannot move between them and are vulnerable (1) • Management change: end of coppicing so woods become uniform and dark; replacement by conifer monocultures (1) • Overgrazing by deer preventing regeneration; introduced pathogens (ash dieback) (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-52; AQA spec 3.1.2.3.4.	
05.1	<i>1 mark for each point.</i> • Irrigation depletes rivers and aquifers (eg Aral Sea, Ogallala); increases evaporation (1) • Soil erosion and compaction reduce water retention and infiltration so runoff and river flow fluctuations increase (1) • Evapotranspiration changes: increased in irrigated arid areas, decreased where forests were cleared (1)	3
05.2	<i>2 marks for each point (partial credit for a partly correct point).</i> • Seasonal / local: avoids energy for heating, lighting greenhouses out of season and long-distance transport (food miles) and the pollution this causes (2) • Organic: uses natural processes (rotation, legumes, manure, biological control, predator habitats) instead of artificial fertilisers and pesticides, so less pollution, energy use and harm to non-target species; free-range and Fairtrade also accepted for welfare / social reasons (2)	4

Question	Marking guidance	Marks
05.3	<i>1 mark for each point.</i> • Food shortages after 1945 (dependence on US food aid); farmers could not afford loans to invest so grants funded machinery, drainage, hedgerow removal, liming, improved livestock (1) • Guaranteed market / price support: government bought surpluses to hold prices up (and sold stores in poor years) giving farmers security to increase output (1) • Consequences: loss of hedgerows and wetlands, intensification, pesticide and fertiliser pollution; later food surpluses (butter mountains) that could not be sold without harming LEDC producers (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 329-333; AQA spec 3.5.1.3-3.5.1.4.	
06.1	<i>1 mark for each point.</i> • As sea otters declined (1960-1980) sea urchins increased (to 350) and kelp area fell (to 25) (1) • As otters recovered after 1990 urchins declined and kelp recovered (1) • Otter and kelp trends are positively correlated; urchins negatively correlated with both (1)	3
06.2	<i>1 mark for each point.</i> • Sea otters are predators of sea urchins so fewer otters allowed the urchin population to rise (1) • Sea urchins graze kelp so the kelp forest was over-grazed and destroyed (1) • Kelp forests are habitat / food for many species including otters (keystone species / trophic cascade); otter recovery restored the balance (1)	3
06.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Over-fishing of sandeels reduced puffins (1) • Over-collection of turtle eggs reduced turtles so jellyfish (their food) increased; over-fishing of sardines / herring (competitors) also increased jellyfish (1) • Loss of pollinating bees (pesticides, wild flower loss) reduces plant reproduction; loss of elephants / large herbivores reduces seed dispersal of trees (1)	2
06.4	<i>1 mark for each point.</i> • They control abiotic features: beaver dams create ponds colonised by aquatic species; elephants create clearings and keep water holes open (1) • Species that depend on these features (light, water, minerals) cannot survive in their absence (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 23-25; AQA spec 3.1.2.2.	
07.1	<i>1 mark for each point.</i> • Energy: low energy-density renewable (solar) / non-renewable fossil fuels (1) • Materials: cyclic - wastes of one process are raw materials of the next / linear - used once then discarded, depletion (1) • Wastes: biodegradable, non-toxic or do not build up / toxic, persistent (eg PCBs, organochlorines, toxic metals) that bioaccumulate (1) • Temperature: low / high (Haber process, incineration) (1)	4

Question	Marking guidance	Marks
07.2	<i>1 mark for each point.</i> - Abundant simple raw materials build into monomers and then a wide range of polymers (carbohydrates, proteins) (1) - They are re-used in cycles so there is no depletion of raw materials or build-up of wastes (1)	2
07.3	<i>1 mark for each point.</i> - Use often disperses the minerals (eg in products spread over the world / corrosion) (1) - Or produces mixtures of materials that are difficult to separate, so sustainable exploitation is reduced (1)	2
07.4	<i>1 mark for each point.</i> - They cannot easily be removed from end-of-life appliances and are not biodegradable (1) - Natural wastes are non-toxic / broken down to raw materials for other processes, whereas persistent toxins may bioaccumulate and biomagnify (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373-374; AQA spec 3.6.2-3.6.3.	
08.1	<i>1 mark for each point.</i> - Moral belief: protecting the environment is the 'right' thing to do regardless of benefit to people (1) - Enlightened self-interest: protecting the environment is done because it causes fewer difficulties (for people) than not doing so (1)	2
08.2	<i>1 mark for each point.</i> - Total impact = population size x the individual (per capita) impact of each person (1) - Per capita impact rises with affluence / access to energy, so both a larger population and a more affluent population increase the total impact (1)	2
08.3	<i>Any 1 from the points below - 1 mark each, maximum 1.</i> - Slash and burn agriculture (1) - Fishing (1) - Discharge of sewage / biodegradable pollutants (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370; AQA spec 3.6.1.	
09.1	<i>1 mark for each point.</i> - Single genes can be transferred from any species, not just closely related ones that produce fertile hybrids (1) - Desired characteristics introduced without linked unwanted ones; faster (1)	2
09.2	<i>1 mark for each point.</i> - Bt: reduces insecticide spraying so fewer non-target deaths, lower cost / energy (1) - Roundup Ready: weeds controlled by spraying glyphosate without harming the crop, so higher yields with less cultivation / tillage (1) - Golden rice: provides vitamin A in the staple diet, reducing about 500 000 cases of blindness a year in India / sub-Saharan Africa (1)	3

Question	Marking guidance	Marks
09.3	1 mark for each point. • Gene transfer by pollen to conventional / organic crops or wild relatives (eg Bt gene killing insects feeding on wild plants) (1) • Antibiotic-resistance marker genes could transfer to pathogenic bacteria; possible allergies; gene transfer to gut bacteria (1) • Detection in the human food chain from crops approved only for animal feed (1) • Economic: seeds must be bought each year from patent-holding companies, raising costs and reducing local crop diversity and seed saving in LEDCs; herbicide-resistant crops may increase herbicide use (1)	4
09.4	1 mark. • Public / political concern that risks are not fully assessed - more evidence needed (precautionary principle) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 323-324; AQA spec 3.5.1.2.	
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • In-situ: species remain in natural conditions with the whole community and inter-species relationships; protected areas (SSSI, NNR, SAC, SPA, Ramsar, MCZ), land ownership, agri-environment schemes, habitat creation and design (area, shape, corridors, diversity), management of abiotic and biotic factors, plagioclimax maintenance, population control, reintroductions and rewilding (Yellowstone wolves, Scottish beavers) • Limits of in-situ: external threats (pollution, climate change), enforcement, cost, conflicts with land use, small fragmented populations • Ex-situ: captive breeding when in-situ is impossible or insufficient; selection criteria; difficulties keeping and breeding (habitat size, food, relationships, breeding triggers, group dynamics, gene pool / inbreeding, hybridisation); techniques (stud books, cryopreservation, AI, embryo transfer, micropropagation, cloning); seed banks; successes (Hawaiian goose, Iberian lynx, red kite, large blue) • Limits of ex-situ: expensive, few species, loss of wild behaviour, small gene pools, release requires suitable habitat and post-release support (hard vs soft release, monitoring) • Judgement: ex-situ is a temporary safeguard / support that only succeeds if in-situ habitat exists to return to; both needed, in-situ fundamental	9

Question	Marking guidance	Marks
10.2	<i>1 mark.</i> Restoring natural processes / communities to a habitat changed by humans, often by reintroducing missing (keystone) species and removing human management such as grazing (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 32-49; AQA spec 3.1.2.3.3-3.1.2.3.4.	
11.1	Discuss how feedback mechanisms, tipping points and diversity influence the stability of natural systems, and how human activities affect them. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) - Sustainability definitions; population and per capita impact; attitudes of rural and urban societies - Negative feedback: cloud cover and albedo, CO₂ and photosynthesis, evaporation and precipitation, density-dependent population regulation - Positive feedback driven by human activities: permafrost, acidification, albedo decline, methane hydrates, fires, cirrus clouds, decomposition - Tipping points: rare under negative feedback; large events (Permian methane hydrate); self-sustaining change from human activities - Diversity and resistance: reefs and rainforests vs low-diversity agroecosystems lacking detritivores, pollinators and predators - Human responses that override feedback (fertilisers, irrigation, pesticides); links to climate change chapter; justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-372; AQA spec 3.6.1.	25

Question	Marking guidance	Marks
11.2	Discuss the extent to which agriculture can be made sustainable while feeding a growing human population. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Need: population possibly 12 billion; humans raise carrying capacity; trophic level of diet • Impacts to be reduced: habitat loss, drainage, nutrient enrichment, introduced species, pesticide and nutrient pollution, nitrate toxicity, greenhouse gases, hydrological changes, soil erosion • Social, economic and political influences: consumer choices, technology availability, grants, guaranteed markets, surpluses, quotas, set-aside, agri-environment schemes • Strategies: cultural and biological pest control, integrated control, organic nutrient supply, legumes, low tillage, water conservation, CWR conservation, renewable energy, methane reduction, habitat retention • Limits: little new land, imports reducing local availability, cost and acceptance of new techniques, past land degradation • Justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-337; AQA spec 3.5.1.3-3.5.1.5.	25